We create chemistry

Safety data sheet

1. Identification

Product identifier

DisPlus 39L

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Performance Chemicals for Oilfield Applications

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Global Oilfield Chemicals

Telephone: +49 621 60-0
E-mail address: RegXcellenceOilfieldChemicals@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Acute Tox. 5 (Inhalation - vapour)
Acute Tox. 4 (oral)
Acute Tox. 4 (dermal)
Skin Corr./Irrit. 2
Eye Dam./Irrit. 1
Aquatic Acute 2
Aquatic Chronic 3

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Danger

Hazard Statement:

H318	Causes serious eye damage.
H315	Causes skin irritation.
H333	May be harmful if inhaled.
H302 + H312	Harmful if swallowed or in contact with skin.
H412	Harmful to aquatic life with long lasting effects.
H401	Toxic to aquatic life.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P273	Avoid release to the environment.
P270	Do not eat, drink or smoke when using this product.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P310	Immediately call a POISON CENTER or physician.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P303 + P352	IF ON SKIN (or hair): Wash with plenty of soap and water.
P361 + P364	Take off immediately all contaminated clothing and wash it before reuse.
P330	Rinse mouth.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

According to UN GHS criteria

Hazard determining component(s) for labelling: (OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO), Alcohols, C13-15, branched and linear, ethoxylated, 2-Butoxyethanol

Other hazards

According to UN GHS criteria

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

Preparation based on: anionic and non-ionic surfactants

Hazardous ingredients (GHS)
According to UN GHS criteria

(OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO)

Content (W/W): >= 25 % - < 50 %	Skin Irrit. 2
CAS Number: 68891-38-3	Eye Dam. 1
	Aquatic Acute 2
	H318, H315, H401
	Specific concentration limit:
	Eye Irrit. 2A: 5 - 10 %
	Eye Dam. 1: > 10 %

Alcohols, C13-15, branched and linear, ethoxylated

Content (W/W): >= 25 % - < 50 %	Acute Tox. 4 (oral)
CAS Number: 157627-86-6	Eye Dam./Irrit. 1
	Aquatic Acute 2
	Aquatic Chronic 3
	H318, H302, H412, H401

2-Butoxyethanol

Content (W/W): >= 20 % - < 25 %	Flam. Liq. 4
CAS Number: 111-76-2	Eye Irrit. 2A
EC-Number: 203-905-0	Acute Tox. 4 (oral)
INDEX-Number: 603-014-00-0	Skin Irrit. 2
	H227, H319, H315, H302

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention. Immediately administer a corticosteroid from a controlled/metered dose inhaler.

On skin contact:
Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:
Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
water spray, dry powder, foam

Special hazards arising from the substance or mixture

harmful vapours, carbon oxides
Generation of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Breathing protection required. Avoid contact with skin and eyes. Avoid inhalation. Ensure adequate ventilation.

Environmental precautions

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up

For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.

7. Handling and Storage

Precautions for safe handling

No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift. Remove contaminated clothing and protective equipment before entering eating areas.

Protection against fire and explosion:
Take precautionary measures against static discharges.

Conditions for safe storage, including any incompatibilities

Segregate from foods and animal feeds. Segregate from strong acids. Segregate from strong bases.
Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

111-76-2: 2-Butoxyethanol
144-55-8: Sodium hydrogencarbonate

Exposure controls

Personal protective equipment

Respiratory protection:
Wear respiratory protection if ventilation is inadequate. Combination filter for gases/vapours of organic compounds and solid and liquid particles (f.e. EN 14387 Type A-P2)

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
nitrile rubber (NBR) - 0.4 mm coating thickness
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing. Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Tightly fitting safety goggles (cage goggles) (e.g. EN 166) and face shield.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Wearing of closed work clothing is required additionally to the stated personal protection equipment. Take off immediately all contaminated clothing. Wash contaminated clothing before reuse. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	liquid	
Form:	liquid	
Colour:	clear	
Odour:	product specific	
Odour threshold:	not determined	
freezing range:	not determined	
boiling temperature:	100 - 101 °C (1.013 hPa)	(OECD Guideline 103)
Flammability:	hardly combustible	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	

Upper explosion limit:		(DIN 51649-1)
	For liquids not relevant for classification and labelling.	
Flash point:	100 - 101 °C	(ISO 2719)
Auto-ignition temperature:	232 °C	
	The product has not been tested. The statement has been derived from the properties of the individual components.	
Information on: 2-Butoxyethanol		
Auto-ignition temperature:	*232 °C*	*(DIN EN 14522)*

Thermal decomposition:	not determined	
pH value:	> 5,5 (100 %(m))	(DIN 19268)
Viscosity, kinematic:	approx. 59 mm2/s (23 °C)	(calculated (from dynamic viscosity))
Viscosity, dynamic:	60 mPa.s (23 °C)	
Solubility in water:	soluble	
Solubility (qualitative) solvent(s):	alcohols soluble	
Partitioning coefficient n-octanol/water (log Kow):	not applicable for mixtures	
Vapour pressure:	62,74 hPa (20 °C)	(OECD Guideline 104)
	71,71 hPa (50 °C)	(OECD Guideline 104)
	155,82 hPa (55 °C)	(OECD Guideline 104)
Density:	1,02 g/cm3 (15 °C)	(ISO 2811-3)
	1,02 g/cm3 (20 °C)	(ISO 2811-3)
	1,00 g/cm3 (50 °C)	
Relative vapour density (air):	not determined	

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: not explosive

Oxidizing properties

Fire promoting properties: not fire-propagating

Self-heating substances and mixtures

Self heating ability: It is not a substance capable of spontaneous heating.

Corrosion to metals

No corrosive effect on metal.

Other safety characteristics

Hygroscopy:	hygroscopic
Evaporation rate:	not determined

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: No corrosive effect on metal.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

No hazardous reactions when stored and handled according to instructions.

Conditions to avoid

Avoid open flames.

Incompatible materials

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products known.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic by inhalation. Of moderate toxicity after short-term skin contact. Of moderate toxicity after single ingestion.

Experimental/calculated data:
ATE (oral): > 300 - < 2.000 mg/kg (calculated)

ATE (by inhalation): 50 mg/l

ATE (dermal): 2.000 mg/kg

Information on: 2-Butoxyethanol
Assessment of acute toxicity:
Of moderate toxicity after single ingestion. The inhalation of a highly enriched/saturated vapor-air-mixture represents an unlikely acute hazard. Virtually nontoxic after a single skin contact.

Information on: (OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO)

Experimental/calculated data:
LD50 rat (oral): > 2.000 mg/kg (BASF-Test)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

LD50 rat (oral): > 2.000 mg/kg (BASF-Test)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: Alcohols, C13-15, branched and linear, ethoxylated

Experimental/calculated data:
LD50 rat (oral): > 300 - 2.000 mg/kg (OECD Guideline 401)
Literature data.

Information on: 2-Butoxyethanol

Experimental/calculated data:
LD50 guinea pig (oral): 1.200 mg/kg (similar to OECD guideline 401)

Information on: (OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO)
Experimental/calculated data:
LD50 rat (dermal): 2.000 mg/kg (OECD Guideline 402)

<u>Irritation</u>

Assessment of irritating effects:
Skin contact causes irritation. May cause severe damage to the eyes.

Information on: (OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO)
Experimental/calculated data:
Skin corrosion/irritation In vitro assay: Irritant. (OECD Guideline 439)

Skin corrosion/irritation rabbit: Irritant. (OECD Guideline 404)

Information on: Alcohols, C13-15, branched and linear, ethoxylated
Experimental/calculated data:

Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)
Literature data.

Information on: 2-Butoxyethanol
Experimental/calculated data:
Skin corrosion/irritation rabbit: Irritant. (BASF-Test)

Information on: (OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO)
Experimental/calculated data:
Serious eye damage/irritation In vitro assay: no irreversible damage (BCOP)

Serious eye damage/irritation rabbit: irreversible damage (OECD Guideline 405)

Information on: Alcohols, C13-15, branched and linear, ethoxylated
Experimental/calculated data:
Serious eye damage/irritation rabbit: irreversible damage (Draize test)
Literature data.

Information on: 2-Butoxyethanol
Experimental/calculated data:
Serious eye damage/irritation rabbit: Irritant. (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:
Based on available data, the classification criteria are not met.

Germ cell mutagenicity

Assessment of mutagenicity:
Based on available data, the classification criteria are not met.

Carcinogenicity

Assessment of carcinogenicity:
Based on available data, the classification criteria are not met.

Reproductive toxicity

Assessment of reproduction toxicity:
Based on available data, the classification criteria are not met.

Developmental toxicity

Assessment of teratogenicity:
Based on the ingredients, there is no suspicion of a teratogenic effect.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
Acutely toxic for aquatic organisms. Harmful to aquatic life with long lasting effects.

Information on: (OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO)
Toxicity to fish:
LC50 (96 h) 7,1 mg/l, Brachydanio rerio (OECD Guideline 203, Flow through.)
The details of the toxic effect relate to the nominal concentration.

Information on: Alcohols, C13-15, branched and linear, ethoxylated
Toxicity to fish:
LC50 (96 h) > 1 - 10 mg/l, Brachydanio rerio (OECD 203; ISO 7346; 84/449/EWG, C.1, static)
Literature data.

Information on: (OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO)
Aquatic invertebrates:
EC50 (48 h) 7,4 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)
The details of the toxic effect relate to the nominal concentration.

Information on: Alcohols, C13-15, branched and linear, ethoxylated
Aquatic invertebrates:
EC50 (48 h) > 1 - 10 mg/l, Daphnia magna (Directive 79/831/EEC)
Literature data.

Information on: (OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO)
Aquatic plants:
No observed effect concentration (72 h) 0,95 mg/l (growth rate), Scenedesmus subspicatus (Algal growth inhibition test)

The statement of the toxic effect relates to the analytically determined concentration.

EC50 (72 h) 27,7 mg/l (growth rate), Desmodesmus subspicatus (OECD Guideline 201)
The statement of the toxic effect relates to the analytically determined concentration.

EC10 (72 h) 4,4 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: Alcohols, C13-15, branched and linear, ethoxylated
Aquatic plants:
EC50 (72 h) > 1 - 10 mg/l (growth rate), Scenedesmus subspicatus (OECD Guideline 201, static)
Literature data.

Information on: (OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO)
Chronic toxicity to fish:
EC10 (28 d) 1,7 mg/l, Pimephales promelas (OECD Guideline 210, Flow through.)
The statement of the toxic effect relates to the analytically determined concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: (OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO)
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d) > 1,19 mg/l, Daphnia magna (OECD Guideline 211, other)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: Alcohols, C13-15, branched and linear, ethoxylated
Chronic toxicity to aquatic invertebrates:
No observed effect concentration > 0,1 - 1 mg/l, Daphnia magna
Literature data.

Persistence and degradability

Information on: (OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO)
Assessment biodegradation and elimination (H2O):
Readily biodegradable (according to OECD criteria).

Information on: 2-Butoxyethanol
Assessment biodegradation and elimination (H2O):
Readily biodegradable (according to OECD criteria).

Bioaccumulative potential

Information on: (OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO)
Assessment bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

Information on: Alcohols, C13-15, branched and linear, ethoxylated
Assessment bioaccumulation potential:
Accumulation in organisms is not to be expected.

Information on: 2-Butoxyethanol
Assessment bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Mobility in soil

Information on: (OLIGOMER) Alcohols, C12-14, ethoxylated, sulfates, sodium salts (> 1 < 2.5 mol EO)
Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: Following exposure to soil, the product trickles away and can - dependant on degradation - be transported to deeper soil areas with larger water loads.Study scientifically not justified.

Information on: 2-Butoxyethanol
Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is not expected.

Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:
The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components.

13. Disposal Considerations

Waste treatment methods

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:
Uncontaminated packaging can be re-used.
Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport
ADN

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

Not applicable

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Acute Tox.	Acute toxicity
Skin Corr./Irrit.	Skin corrosion/irritation
Eye Dam./Irrit.	Serious eye damage/eye irritation
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Skin Irrit.	Skin irritation
Eye Dam.	Serious eye damage
Flam. Liq.	Flammable liquids
Eye Irrit.	Eye irritation
H318	Causes serious eye damage.
H315	Causes skin irritation.

H401	Toxic to aquatic life.
H302	Harmful if swallowed.
H412	Harmful to aquatic life with long lasting effects.
H227	Combustible liquid.
H319	Causes serious eye irritation.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.